

Lab 1 Related Exercise

Lab 1

Understanding Basics of Machine Learning Programming using Python – Pandas & Numpy

Aim: To understand the fundamentals of Python programming for Machine Learning, and to explore the NumPy and Pandas libraries for numerical computation and data handling.

Theory / Description

Every Machine Learning workflow begins with loading, exploring and manipulating data, and Python has become the de-facto language for this because of two foundational libraries.

Numpy

This is a quick overview of arrays in NumPy. It demonstrates how n-dimensional ($n \geq 2$) arrays are represented and can be manipulated. In particular, if you don't know how to apply common functions to n-dimensional arrays (without using for-loops), or if you want to understand axis and shape properties for n-dimensional arrays.

Learning Objectives

You should be able to:

- Understand the difference between one-, two- and n-dimensional arrays in NumPy;
- Understand how to apply some linear algebra operations to n-dimensional arrays without using for-loops;
- Understand axis and shape properties for n-dimensional arrays.

The Basics

NumPy's main object is the homogeneous multidimensional array. It is a table of elements (usually numbers), all of the same type, indexed by a tuple of non-negative integers. In NumPy dimensions are called axes.

For example, the array for the coordinates of a point in 3D space, `[1, 2, 1]`, has one axis. That axis has 3 elements in it, so we say it has a length of 3. In the example pictured below, the array has 2 axes. The first axis has a length of 2, the second axis has a length of 3.

```
[[1., 0., 0.],  
 [0., 1., 2.]]
```

NumPy's array class is called `ndarray`. It is also known by the alias `array`. Note that `numpy.array` is not the same as the Standard Python Library class `array.array`, which only handles one-dimensional arrays and offers less functionality. The more important attributes of an `ndarray` object are:

ndarray.ndim

The number of axes (dimensions) of the array.

ndarray.shape

The dimensions of the array. This is a tuple of integers indicating the size of the array in each dimension. For a matrix with n rows and m columns, shape will be (n,m) . The length of the shape tuple is therefore the number of axes, `ndim`.

ndarray.size

The total number of elements of the array. This is equal to the product of the elements of shape.

ndarray.dtype

An object describing the type of the elements in the array. One can create or specify dtype's using standard Python types. Additionally NumPy provides types of its own. `numpy.int32`, `numpy.int16`, and `numpy.float64` are some examples.

ndarray.itemsize

The size in bytes of each element of the array. For example, an array of elements of type `float64` has `itemsize` 8 ($=64/8$), while one of type `complex32` has `itemsize` 4 ($=32/8$). It is equivalent to

`ndarray.dtype.itemsize`. **`ndarray.data`**

The buffer containing the actual elements of the array. Normally, we won't need to use this attribute because we will access the elements in an array using indexing facilities.

Pandas

Pandas is a Python library used for working with data sets. It has functions for analyzing, cleaning, exploring, and manipulating data. The name "Pandas" has a reference to both "Panel Data", and "Python Data Analysis" and was created by Wes McKinney in 2008.

Use of Pandas

Pandas allows us to analyze big data and make conclusions based on statistical theories. Pandas can clean messy data sets, and make them readable and relevant. Relevant data is very important in data science.

Data Science: is a branch of computer science where we study how to store, use and analyze data for deriving information from it.

Pandas gives you answers about the data. Like:

- Is there a correlation between two or more columns?
- What is average value?
- Max value?
- Min value?

Pandas are also able to delete rows that are not relevant, or contains wrong values, like empty or NULL values. This is called *cleaning* the data.

Once Pandas is installed, import it in your applications by adding the import keyword:

Pandas is built on top of NumPy and introduces two labelled data structures: the Series (a 1-D labelled array, essentially one column) and the DataFrame (a 2-D labelled table, like a spreadsheet or SQL table). Pandas makes it easy to read files (CSV, Excel, SQL), inspect data (`head()`, `info()`, `describe()`), select/filter rows and columns, handle missing values, and group and aggregate data. Together, mastering NumPy and Pandas is the essential first step before any regression, classification or clustering algorithm can even be applied, because

every model in Scikit-learn ultimately expects a clean NumPy array or Pandas DataFrame as its input.

Solved Example:

Q1. Create a NumPy array of integers from 1 to 20, reshape it into a 4x5 matrix, and find its transpose, sum, mean, and standard deviation.

Q2. Write a NumPy program to create two 3x3 matrices and perform addition, subtraction, multiplication (element-wise and matrix multiplication) and find the determinant of one matrix.

Q3. Create a Pandas DataFrame from a dictionary containing student names, marks in three subjects, and display the top rows, data types, and summary statistics.

Problem: A college wants to quickly analyze the marks of 6 students in three subjects (Maths, Physics, Chemistry) stored as a Pandas DataFrame — find each student's total and average marks, identify the topper, and compute subject-wise class averages.

Problem : Handle missing values in a DataFrame using Pandas (identification, dropping and imputation).

Exercise Questions

1. Convert a 1-D array into a 2-D array with 3 rows.

Start with:

Assign-1 = np.array([0, 1, 2, 3, 4,
5, 6, 7, 8]) Desired output:

[[0, 1, 2]

[3, 4, 5]

[6, 7, 8]]

2. Write a Python program to demonstrate basic operations using NumPy arrays (creation, reshaping, indexing, slicing).

3. Consider the following matrices :

A = ((1, 2, 3), (4, 5, 6), (7, 8, 10)) and B = ((7, 8, 10), (4, 5, 6), (1, 2, 3))

Write a python program to perform the following using Numeric Python (numpy).

- i) Add and Subtract of the Matrix A and B, print the resultant matrix C for add and E for subtract.
- ii) Compute the sum of all elements of Matrix A, sum of each column of Matrix B and sum of each row of Matrix C
- iii) Product of two matrices A and B, and print the resultant matrix D
- iv) Sort the elements of resultant matrix C and print the resultant Matrix E.
- v) Transpose the Matrix E and print the result

- 4. Write a Python program to group a dataset by a categorical column and compute aggregate statistics (mean, sum, count) for each group using `groupby()`.
- 5. Write a Python program to pivot a dataset into a cross-tabulated summary table using `pivot_table()`, with a specified aggregation function.
- 6. Write a Python program to apply a custom function row-wise/column-wise on a DataFrame using `apply()` and lambda expressions (e.g., creating a new derived feature).
- 7. Write a Python program to detect and remove duplicate rows from a dataset, and identify duplicate values based on a subset of columns.
- 8. Write a Python program to filter rows using multiple conditions combined with `.loc[]` and boolean indexing (e.g., rows where `feature1 > threshold AND category == 'X'`).
- 9. Write a Python program to rank rows within groups using `groupby().rank()` (e.g., ranking students by marks within each class).

- 10. *A logistics company stores the daily number of parcels delivered by 4 warehouses over a 5-day week as a grid of numbers. They want to convert this into a NumPy matrix to quickly compute warehouse-wise and day-wise totals.*

Task: Represent the data as a NumPy 2-D array (4 warehouses x 5 days), then compute the total parcels per warehouse (row-wise) and per day (column-wise), and the grand total.

- 11. An HR analytics team receives an employee dataset with some missing 'Years of Experience' values and duplicate entries caused by a data entry error, and needs it cleaned before it can be used for a salary-prediction model.

Task: Load the employee data into a Pandas DataFrame, remove duplicate rows, and fill missing 'Experience' values with the column median.

- 12. *A retail chain wants a quick report of total and average monthly sales for each of its store branches, grouped from a raw transaction-level DataFrame.*

Task: Given a DataFrame of individual transactions with Branch and Sales columns, use `groupby()` to compute the total and average sales per branch, sorted from highest to lowest total sales.

